

Reading, speaking, and listening

- The ease with which people can read, listen, or speak differs
 - Many prefer listening to reading
 - Reading can be quicker than speaking or listening
 - Listening requires less cognitive effort than reading or speaking
 - Dyslexics have difficulties understanding and recognizing written words

Applications

- Speech-recognition systems allow users to interact with them by asking questions
 - e.g. Google Voice, Siri
- Speech-output systems use artificially generated speech
 - e.g. written-text-to-speech systems for the blind
- Natural-language systems enable users to type in questions and give text-based responses
 - e.g. Ask search engine

Design implications

- Speech-based menus and instructions should be short
- Accentuate the intonation of artificially generated speech voices
 - they are harder to understand than human voices
- Provide opportunities for making text large on a screen